



FEASIBILITY STUDY

Dorm Dash



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Executive Summary

The North-West University (NWU) is facing more and more challenges when handling more than 13,000 residence beds on three campuses, as existing systems are failing to fulfil operational and communication needs. Current tools, such as the outdated VSS system and StarRez, have restrictions in real-time occupancy monitoring, allocation effectiveness, and adaptation to the specific cultural and administrative requirements of South African universities.

This proposal details the creation of a web-based residence management platform by the name of Dorm Dash as an internal project guided by faculty, driven by students. The system seeks to simplify room assignments, boost communication between residence personnel and students, and increase student involvement through elements such as event management, group collaboration, and in-system voting.

Dorm Dash shows a big chance for commercial growth through NWU Enterprises. It gives a white-labelled fix that can fit other places and work fields. This plan fits NWU's innovation agenda, aids in skill build, and may cut costs and bring in new revenue streams.

The initiative corresponds with NWU's agenda for innovation and modernization, providing both immediate institutional advantages and opportunities for commercialization via NWU Enterprises. A staged Minimum Viable Product (MVP) will concentrate on essential features like role-specific dashboards, live room oversight, and event notifications. This method guarantees practicality within academic limits while allowing for expansion in the future. The initiative tackles existing operational inefficiencies while also establishing NWU as a technology leader that can provide tailored solutions for higher education.

This feasibility study evaluates looks at three ways to fix NWU's home manage issues: developing Dorm Dash in-house, retaining the current system, or procuring a new vendor solution. Dorm Dash shows the best long hold fit, with high ease to change, with high adaptability, and the chance to be commercialised it. While it requires more initial effort, careful phasing and faculty mentorship can mitigate risks related to time, resources, and implementation complexity.

Problem Statement

Business Problem

North-West University (NWU) currently faces significant challenges in managing its student housing effectively. Across its three campuses, the institution oversees more than 13,000 residence beds, yet the systems in place are outdated, fragmented, and not designed for the specific demands of residence management.

At present, NWU relies on a combination of its general Student Information System and a legacy VSS administrative tool. While these systems fulfil certain administrative functions, they are not purpose-built for residence operations. As a result, several persistent issues have emerged:

- **Inefficient allocation processes** – Assigning students to rooms involves manual, time-consuming steps that make it difficult to ensure fairness and transparency without specialised tools.

- **Lack of real-time occupancy tracking** – There is no quick way to determine whether a bed is vacant, reserved, occupied, or under maintenance, leading to underutilisation of available space.
- **Limited reporting capabilities** – Residence staff struggle to generate detailed, accurate reports to inform decision-making, often having to piece together data from multiple sources manually.
- **Inconsistent processes across campuses** – Each campus operates differently, without a unified platform to standardise procedures, creating inefficiencies and misalignment.
- **Suboptimal student experience** – Students face an outdated and inconvenient process for applying for housing or checking application status, which can lower satisfaction and negatively affect NWU's image.

Business Opportunities

The challenges outlined above also present NWU with a strategic opportunity to reimagine its residence management approach and create long-term value for both the institution and its stakeholders:

- **Customised solution for NWU's needs** – Developing an in-house Residence Management Platform would allow NWU to build a system aligned with its operational processes, policies, and cultural context, offering greater flexibility than generic commercial software.
- **New revenue potential** – A successful in-house platform could be commercialised and offered to other universities through NWU Enterprises, generating an additional income stream while enhancing NWU's profile as an innovator in higher education.
- **Enhanced service delivery and reputation** – A streamlined, transparent, and user-friendly housing process would improve student satisfaction and retention, strengthening NWU's reputation locally and nationally.
- **Institutional innovation model** – This project could serve as a model for future student-driven initiatives, fostering a culture of innovation, cost savings, and continuous improvement.
- **Skills development for students** – Involving students in the system's development would provide practical, real-world experience in designing, building, and implementing large-scale solutions, improving graduate employability.
- **Cost savings and operational control** – By owning and controlling its own system, NWU can reduce reliance on expensive third-party vendors, implement updates as needed, and adapt quickly to operational changes.

The opportunity extends beyond solving immediate operational challenges, it positions NWU as a leader in developing innovative, home-grown solutions for higher education, while simultaneously improving internal efficiency, student satisfaction, and institutional competitiveness.

Requirements Statement

Business Drivers

A number of factors must be considered for the best decision to be made with regards to the way forward with residence management:

- **Enhanced Student Experience** :Offer an intuitive platform for applying for accommodation, checking status, submitting maintenance requests, and accessing participation marks.
- **Resource Optimization**: Monitor and optimize space usage, utilities, and maintenance schedules to reduce costs and wastage.
- **Improved Communication**: provide a means of direct and timely communication between residence staff and students with certain topics like when to anticipate a team being sent over to fix something that broke

Business Requirements

The various business opportunities and problems along with the system requirements needed to combat these issues will be displayed in the table below

Business Problem/ Opportunity	Project System Requirements
Students ability to improve their set of skills	The project can serve as a practical training ground for students, helping them gain real-world experience.
Difference in students availability	Student developers and users have varying schedules and availability.
Strict financial boundaries	The student creating the system are limited to using cost-free platforms to create the system
Current system is behind on maintaining records of what facilities are needed	The system must be able to implement a facilities management module for logging, tracking, and prioritising maintenance requests.

Feasibility Assessment

In higher education, where institutional priorities often intersect with budgetary constraints, the decision to adopt, adapt, or develop a new system involves trade-offs that extend beyond purely operational concerns.

In this section we will compare three potential pathways: the in-house development of Dorm Dash, the continuation of current systems, and the adoption of an alternative vendor-operated platform. Each option is examined against multiple feasibility criteria which includes technical, financial, operational, and human-resources. The aim is to position NWU's decision within a broader understanding of higher-education technology adoption, where feasibility is as much about strategic positioning and institutional resilience as it is about meeting today's operational needs.

Solution 1: Dorm Dash

Description

Dorm Dash is a web-based platform built for managing student residences at the NWU. It would be developed in-house by student teams with guidance from faculty. The system brings together key functions in one place. These include tracking room use in real time, sending updates to residents, managing events, and logging maintenance needs. The aim is to replace outdated tools with something tailored to the NWU's way of working. The project can start small, with core features, and grow over time. It is also designed so that it could be adapted for other universities in the future, creating a possible income stream for the NWU.

Drawbacks

Building Dorm Dash will not be without challenges. Student developers may face pressure from their academic workload. This could slow the pace of work or affect quality. Some features, like real-time tracking across all campuses, may be harder to deliver than expected. New systems can also meet resistance from staff and students who are used to the old way of doing things. There will be costs for servers, testing, and ongoing support, even if there are no licence fees. Without strong planning and ongoing backing from the university, these issues could delay the project or reduce its long-term success.

Benefits

High institutional fit, as the system is purpose-built for NWU's operational, cultural, and administrative needs. It's also customised to South African higher education, it directly addresses residence management challenges such as real-time bed tracking, unified communication, and streamlined processes across campuses. This improves operational efficiency, reduces manual work, and supports accurate data-driven decisions.

It also enhances student engagement by providing a user-friendly platform for accessing residence details, RSVPs, polls, and communication tools. It has strong alignment with students, residence staff, and management ensures broad support. Additionally, the system holds commercial potential for other HEIs facing similar challenges, positioning NWU as a sector innovator. With in-house development, long-term costs are significantly lower than international solutions, making it both strategically aligned and financially sustainable.

Feasibility Scores

Criterion	Number	Explanation
User Friendly	4	The system should be easy to use for student leaders as well as staff in charge of residences
Budget Consciousness	5	The system will be cheaper seeing as it is to be created by students at the university
Implementation	2	The system's implementation has a higher potential of being harder to implement
Time Consciousness	3	The system's creation and implementation will take a serious amount of time

Adaptability	4	One of the system's minimum requirements is to be adaptable to other topics
End User Instruction	4	The system is to be made to be easy to teach students as well as staff
Human Resources	3	The people involved in the creation and implementation are students, this saves money on labour costs

Risks

Time Consciousness

Student teams will have to balance development work with their studies. Busy exam periods or project deadlines in other courses could slow progress. If timelines are not realistic, there is a risk of rushing features, which could lead to errors or incomplete functions. A longer build time could also mean that early enthusiasm fades before the system is fully launched. Scope creep is also a major issue.

Human Resources

Although using students reduces labour costs, it also means the team will have less long-term stability. Students graduate or move on, and this can leave knowledge gaps. If a key developer leaves mid-project, replacing their skills may be difficult. This makes good documentation and mentoring essential to keep the project on track.

Implementation

Rolling out a new system across multiple campuses will not be simple. Data from old systems must be moved over without loss or corruption. Staff and student training will be needed so everyone can use the new tools effectively. Technical issues during rollout could frustrate users and harm confidence in the platform. Without careful planning, these hurdles could slow adoption and create a poor first impression.

Estimated Cost Breakdown

Development and Technical Setup

Item	Estimated Cost	Notes
Cloud hosting (1 year)	R12000	Based on mid-tier AWS plan for web application and database hosting
Domain registration	R120	Annual fee for .co.za or .ac.za domain
SSL certificate	Free	Free by using Cloudflare with auto renewal
Subtotal	R12120	

Implementation and Data Migration

Item	Estimated Cost	Notes
Database consultancy	R7500	Expert input for performance/security optimisation
Backup and recovery setup	R2500	Cloud backups and redundancy systems

Subtotal	R10000
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Training and Change Management

Item	Estimated Cost	Notes
Staff training sessions	R3000	In-person workshops or online by student assistants
Training materials	R2000	Printed guides/manuals or digital
Onboarding support (3 months)	R6000	Student assistants to help users transition
Subtotal	R11000	

Total

Category	Price
Development and Technical Setup	R12120
Implementation and Data Migration	R10000
Training and Change Management	R11000
Contingency (10%)	R3120
Grand total	R36240

Conclusion

Dorm Dash offers NWU a system built for its own needs rather than forcing existing processes to fit an off-the-shelf product. It can improve how residences are managed, make communication faster, and give students a better experience. The project also has the potential to be adapted for other universities, creating a possible income stream. The first-year costs are lower than most vendor options, and the university will own the platform outright. With the right commitment and guidance, Dorm Dash can become both a practical solution for NWU and a long-term asset.

Solution 2 : Maintain Current System

Description

Maintaining the current residence management setup means continuing to use the existing combination of legacy software and manual processes already in place at NWU. This option avoids the immediate effort and cost of developing or buying a new system. It keeps staff working within tools they already know, which can make daily operations feel familiar and predictable. But it also means living with the same limitations that have caused problems in the past. Any efficiency gains would be minimal and opportunities for innovation or long-term cost savings would be lost. This path is the least disruptive in the short term, but it risks leaving the university behind as technology and expectations continue to change.

Drawbacks

Keeping the current system in place means that existing inefficiencies will remain part of daily operations. Staff will still spend extra time on manual tasks that could be automated, such as piecing together reports from multiple sources or tracking room availability by hand. The lack of real-time occupancy tracking would continue to limit how well space is used. Beds might remain vacant while demand is high, simply because the information is not immediately available.

There is also the risk of higher hidden costs. While the system is already paid for, ongoing maintenance and occasional workarounds can still consume time and money, the StarRez proposition forms part of these hidden costs to adapt to the NWU's current needs and culture. If technical problems arise, fixing them in an outdated system may be more difficult and more expensive than expected.

Benefits

Sticking with the current system has the advantage of being the least disruptive choice in the short term. Staff are already familiar with the tools and processes, so there is no need for large-scale training or a long adjustment period. Day-to-day operations could continue without the interruptions that often come with implementing new technology.

This option also avoids significant upfront costs. Since the system is already in place, the university would not need to invest in development, licensing or new infrastructure. For a period of time, this can make it appear to be the most budget-friendly choice.

Feasibility Scores

Criterion	Number	Explanation
User Friendly	3	There are already staff that know how to use them.
Budget Consciousness	2	This is the financially fair option seeing as cost of operating it is already pre-empted
Implementation	3	The system would not take much as it already works
Time Consciousness	5	No time taken to implement because it has long since been in use
Adaptability	1	Not very adaptable as there are many lapses in the system
End User Instruction	3	None needed as there is no new functionality
Human Resources	5	In leaving the system as is, this frees up the workload on the staff

Risks

Operational inefficiencies

Manual processes and workarounds may seem manageable now, but they can quickly become bottlenecks during busy periods. Over time, this could lead to delays, errors, and increased frustration among staff and students.

Security vulnerabilities

Older systems often lack modern security measures. This raises the risk of data breaches or unauthorised access to sensitive student and housing information. Any incident could damage the university's reputation and result in costly recovery measures.

Ageing technology

As the system continues to age, it will become harder to maintain and update. Compatibility issues with newer hardware or software could make future upgrades or integrations more expensive and more complex to implement.

Estimated Cost Breakdown

Development and Technical Setup

Item	Estimated Cost	Notes
System hosting and server costs	R8000	Annual hosting fees for the existing system and database
Licence renewals or vendor support fees	R10000	Annual renewal costs for proprietary software licences and vendor support contracts
Security updates and patches	R2000	Periodic updates to keep the system secure
Minor software enhancements	R2000	Small changes to improve usability or fix issues
Subtotal	R12000	

Implementation and Data Migration

Item	Estimated Cost	Notes
Legacy data maintenance	R4000	Regular checks to ensure data accuracy and integrity
Subtotal	R4000	

Training and Change Management

Item	Estimated Cost	Notes
New staff onboarding	R3000	Time and resources for training new staff in the existing system
Refresher sessions for current staff	R1500	Occasional training to keep staff updated on minor changes
Process documentation updates	R1000	Keeping manuals and guides current with any process changes
Subtotal	R5500	

Total

Category	Price
Development and Technical Setup	R12000
Implementation and Data Migration	R4000

Training and Change Management	R5500
Contingency (10%)	R2150
Grand total	R23650

Conclusion

Keeping the current system offers short-term stability and avoids the disruption of a major technology change. Staff can continue working with tools they already know, and upfront costs remain low compared to developing or purchasing a new platform. However, this stability comes at the cost of long-term efficiency and innovation. While it may seem like the safest choice now, it risks leaving NWU further behind in meeting modern expectations for residence management.

Solution 3: Other Vendors

Description

Choosing an external vendor means adopting a ready-made residence management system developed and supported by a third party. These platforms are usually built for a wide range of institutions and come with established features, technical support and regular updates. Implementation can be faster than building a system from scratch, and the learning curve may be gentler when vendor training is included. This route also requires NWU to adapt its processes to fit the software rather than tailoring the software to fit NWU.

Most importantly, it creates a dependency on the vendor for future changes, and licensing or subscription fees can become a significant long-term expense. While it offers a quicker path to a modern system, it limits the university's control over how that system evolves.

Drawbacks

Relying on an external vendor means NWU will have less control over the system's development and future changes. Any new features or adjustments must be requested through the vendor, which can be slow and sometimes costly. This can make it harder to adapt the system to meet local requirements or respond quickly to operational changes. The cost of licensing and ongoing subscriptions is another concern. While these systems are ready to use, their annual fees can be high, especially for a university managing thousands of residence beds. Over time, the total cost may exceed that of building and owning a system in-house.

Cultural and administrative fit can also be a challenge. Many vendor solutions are designed for international markets, and may not fully reflect South African university processes or policies. Adapting them can require extra configuration or customisation, which adds to the expense..

Benefits

Working with an established vendor offers the advantage of a system that is already built, tested, and in use at other institutions. This means NWU could move to a modern platform much faster than if it developed one internally. The initial setup process is often guided by the vendor, reducing the technical workload on the university's own staff. Vendor solutions usually come with professional support services. Staff training is often part of the package, giving users the chance to learn the system quickly and with expert guidance.

These systems also benefit from a track record of use in real-world settings. Features are typically refined over time based on feedback from many clients, which can result in a stable and reliable product. For a university seeking quick improvements to residence management without a lengthy development phase, this approach offers a practical and relatively low-risk way to modernise..

Feasibility Scores

Criterion	Number	Explanation
User Friendly	1	In as much as they have been approved for industry, most of the users might be confused on how to operate the system
Budget Consciousness	1	The system has a high probability of being more expensive than the first two options
Implementation	4	The industry approved system means the system is easier to put into place
Time Consciousness	3	It will be relatively quicker to implement a ready-made system
Adaptability	2	Possibility of challenges in attempts to scale up the system
End User Instruction	3	The existing systems would be easier for new users to operate as they have been approved for the business requirements
Expenditure of Human Resources	3	It will be cheaper seeing as the people need only be there to assist in downloading the system at most

Risks

High ongoing costs

Vendor licensing and subscription fees can be substantial. While the system may be quick to implement, these costs will continue year after year. Over time, they may exceed the cost of developing and maintaining an in-house system.

Limited flexibility

The vendor controls the system's features and roadmap. If NWU needs changes or new functionality, it must rely on the vendor's timeline and priorities. This can slow innovation and make it harder to tailor the system to local needs.

Cultural and operational misfit

Many vendor solutions are built for international markets. Processes, terminology, or compliance requirements may not fully match the South African higher education environment. Adapting the system to fit NWU could require extra configuration or compromise on certain workflows.

Vendor dependency

Once implemented, the university becomes dependent on the vendor for updates, support, and future compatibility. If the vendor changes pricing, reduces support, or discontinues the product, NWU may face costly and disruptive transitions.

Estimated Cost Breakdown

Development and Technical Setup

Item	Estimated Cost	Notes
Initial licensing or purchase fee	R180000	One-time payment or first-year licence cost for a large-scale vendor solution
Setup and configuration	R35000	Vendor service to customise basic branding, user roles, and workflows
Subtotal	R215000	

Implementation and Data Migration

Item	Estimated Cost	Notes
Data migration services	R25000	Vendor-led migration of all existing residence and student data
Integration with other NWU systems	R20000	Linking to student information, finance, and facilities systems
Subtotal	R45000	

Training and Change Management

Item	Estimated Cost	Notes
Staff training sessions	R10000	In-person or online workshops by student assistants
Training materials	R8000	Printed or digital guides/manuals
Onboarding support (3 months)	R12000	Student assistants to help users transition
Subtotal	R30000	

Total

Category	Price
Development and Technical Setup	R215000
Implementation and Data Migration	R45000
Training and Change Management	R30000
Contingency (10%)	R29000
Grand total	R319000

Conclusion

An external vendor solution would give NWU quick access to a modern, fully supported residence management system. The implementation process would be faster than building a platform internally, and professional support could reduce technical risks during rollout. This convenience comes with a high ongoing cost and limited flexibility for adapting the system to NWU's specific needs. The university would be dependent on the vendor for updates, pricing, and feature changes, which could limit long-term control. While this option delivers immediate functionality and stability, it is the most expensive route and offers the least opportunity for innovation or future customisation.

Feasibility Ranking

The solutions feasibility will be determined according to the following to how well they match the following characteristics:

Criterion	Dorm Dash	Do Nothing	Other System	Vendor
User Friendly	4	3	1	
Budget Conscience	5	2	1	
Implementation	2	4	3	
Time Consciousness	3	5	4	
Adaptability	4	1	2	
End User Instruction	4	3	3	
Human Resources	3	5	3	
Total	25	23	17	

Feasibility Results

This analysis evaluates whether the Residence Management Platform (RMP) can be successfully developed within NWU's academic and resource constraints. Two solutions are compared:

Solution 1 (Dorm Dash): A student built, in house system.

In terms of technical feasibility, the system uses modern technology (React, Node.js, MySQL), fits NWU's needs, students can build it with mentorship, however the students may need support in the form of faculty oversight. It will be cheaper to have the student create the system; however, it may be hard on the students themselves when juggling academics as well as creating the system, time will be a serious constraint.

Solution 2 (Current System): Existing vendor solution

This solution involves leaving the current system in place, in as much as it is easier to leave it be, it causes more issues as how the system is as right now. It is also more expensive to rely on the one that exists. In using this system, there is no opportunity to add students in residences and keep track of the facilities they need. It will be much easier and quicker to leave the system be.

Solution 3 (New Vendor-operated System): New vendor solution

This involves going to another vendor for a more robust and easier to implement system. It has the added benefits of being easy to implement and easy to use with ready-made resources, it has the downsides of it are the higher costs to buy it and has no possibility of being made scalable without the vendors permission.

Recommendation

The assessment of the three options shows that Dorm Dash is the most suitable path for NWU. It offers the chance to design a system that works for the university's processes and culture. It can be built in stages so that the most important features are delivered first. Later additions can follow as time and resources allow. While keeping the current system would cause the least short-term disruption, it would also keep existing problems in place. Staff would continue to face slow processes and manual work. Over time, this would make it harder for NWU to stay competitive in residence management.

The vendor solution would be fast to set up and would come with professional support. It would create high ongoing costs and reduce the university's ability to adapt the system to its own needs. For these reasons, the recommendation is to move forward with Dorm Dash. Faculty oversight should be in place from the start. Student workloads should be monitored so that deadlines remain realistic. By doing this, NWU can manage risks while building a system that will serve the university well into the future.

Conclusion

Dorm Dash provides the best balance of cost, control, and growth potential. It can be shaped to fit NWU without the limits of an off-the-shelf product. It also opens the door to new opportunities, including adapting the system for use by other institutions.

With careful planning, strong guidance, and a phased rollout, the risks linked to student-led development can be kept to a minimum. The alternative options may be easier in the short term, but they either fail to address current issues or create high long-term expenses. Choosing Dorm Dash is an investment in both immediate improvement and long-term capability. It is a step that will modernise residence management and position NWU as a leader in higher education innovation.